🚀 Lecture 04.13: Practice Problems (Part a & b) — Common Elements & Frequencies

📌 Executive Architectural Summary

This lecture tackles hands-on algorithmic problems synthesizing lists, sets, and dictionaries: finding common elements across collections using set intersection, and computing word frequencies using dictionary hash tables.

🔬 Core Engineering Mechanics

1. Problem A: Common Elements Extraction

Given two lists L_1 and L_2, find all elements present in both:

$$extCommon = extlist(extset(L_1) \cap extset(L_2))$$

Time Complexity: $\mathcal{O}(|L_1| + |L_2|)$ compared to naive nested loops $\mathcal{O}(|L_1|imes|L_2|)$.

2. Problem B: Word Frequency Histogram

Given a corpus string, count the frequency of each unique word using a dictionary accumulator pattern.

💻 Production Python Implementation

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# Part A: Common Elements via Set Intersection
list1 = [1, 2, 2, 3, 4, 5]
list2 = [2, 4, 6, 8]
common = list(set(list1) & set(list2))
print("Common Elements:", sorted(common))

# Part B: Word Frequency Counter
sentence = "python is great and python is fast and python is dynamic"
word_counts = {}
for word in sentence.split():
    word_counts[word] = word_counts.get(word, 0) + 1
```

```
print("
Word Frequencies:")
for w, c in word_counts.items():
    print(f"'{w}': {c}")
```